One Winding per Sheet

Auditing and correcting spiral-fit windings of Herculaneum scrolls against the surface prediction and the CT

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Vesuvius Challenge, Progress Prize submission, October 2026 · code (MIT): github.com/TAUIL-Abd-Elilah/spiral-winding-audit · developed with AI assistance under the author's direction

Abstract. The spiral-fit recipe that produced the PHerc. 343 First Letters (villa `fit_spiral`, a nearest-sheet snap onto the m7 surface prediction, lasagna flattening) is now the standard way to search the remaining scrolls, and its windings are accepted by eye. We measure them per node. On four bands of three scrolls (PHerc. 0175A, 0813, 0800; 120 windings each), 17–24 % of the flattened nodes lie within 0.3 sheet pitch of the next winding, and 15–34 % lie on no predicted sheet. Two independent CT checks confirm it: node pairs that the audit puts on one sheet render the same papyrus texture (correlation 0.55 against 0.28 for adjacent wraps). Counting along radial rays, the fits place 1.2–1.6 windings on each predicted sheet: `fit_spiral`'s spiral pitch stays near its starting value of 16 voxels while the predicted sheets are 22–29 voxels apart radially, and the snap then pairs windings on shared sheets. Against bright layers counted in the raw CT the excess is smaller (0.9–1.3 windings per CT layer on PHerc. 0800, 1.1–1.5 on PHerc. 0175A): part of it is sheets the prediction misses. The winning PHerc. 343 pair shows the same on 21 % of a winding, mostly outside its letters. We release CPU tools that measure this before any rendering (`density`, `audit`), repair what a re-snap can repair (`repair`, checked against the CT), and mask the single-sheet surface that is read once (`pieces`). The same 1.5 windings per sheet appears in the fit's own spiral space, where the sheets' winding phase is nearly unrelated to the windings (alignment 0.11–0.15); we add this check to villa's spiral-fitting tools. Either way the rendered windings duplicate papyrus, and every fit should be checked, against the prediction and the raw CT, before rendering; whether the remedy is fewer windings or better prediction recall depends on the scroll.

1. Problem

A scroll is a rolled sheet. Virtual unwrapping must produce, for every wrap, a surface that stays on that wrap's papyrus all the way round, and no surface that runs on a neighbour's papyrus. The Vesuvius Challenge lists sheet switches and mergers among its main bottlenecks, and its judging criteria for unwrapping ask that output surfaces "don't jump across sheets in cross-section".

Since the PHerc. 343 award in October 2026, several groups fit a global spiral to a scroll band with villa's `fit_spiral`, export one surface per winding, snap each node along its normal to the nearest sheet of the published m7 surface prediction within ±15 voxels, flatten with lasagna, render and run ink detection. Each step is checked by eye on a few windings. Nothing checks that winding k and winding k+1 lie on different sheets everywhere, or how many sheets a fit has per winding. This work measures both, explains the failure, and gives tools that find it before GPU time is spent on rendering and ink.

Terminology: "sheet" here means one turn's papyrus as the surface prediction marks it. Villa's SCROLL_ANATOMY.md calls that the papyrus of a winding and reserves "sheet" for a manufactured kollema; the villa tool therefore says "layer".

2. Data

scroll / band	scan	windings	stages audited	CT used
PHerc. 0175A, z 6900–7900	8.64 µm, 116 keV	w010–w129	fit, snap, flatten, repair	66-layer renders of w070, w073, w092
PHerc. 0813, z 12200–13200	9.36 µm	w010–w129	fit, snap, flatten	–
PHerc. 0800, z 5300–6300	8.64 µm	w010–w129	fit, snap, flatten	cached CT chunks (3 windings)
PHerc. 0800, z 6300–7800	8.64 µm	w010–w129	fit, snap	22-layer renders of w095–w097; cached CT (3 windings)
PHerc. 343 (control)	8.64 µm	w047, w048 (released by E. Nieuwlaar)	flatten	66-layer render of the letter image's mesh

All fits use villa `fit_spiral` at commit `0fb38c45` with 20k steps and the snap of the PHerc. 343 recipe; the surface prediction is the published m7 store of each volume (level 0, threshold 128). Windings are tifxyz grids with a 20-voxel node spacing.

3. Methods

3.1 Distance to the next winding's surface

For each node of winding k, the nearest node of winding k+1 at a similar angle around the turn is found with a KD-tree that carries the angular position as a fourth coordinate, so the spiral seam (where the end of one winding meets the start of the next on the same sheet) is not read as a merger. The node is then projected onto the tangent plane of k+1 at that node. Measuring to the nearest *node* instead misses windings that lie on top of each other, because on a 20-voxel grid the nearest node can be 10 or more voxels to the side. Against exact point-to-triangle distances to the neighbouring mesh the plane distance agrees within one voxel for about 90 % of nodes.

3.2 Sheet pitch from the prediction

The pitch that the thresholds refer to is the median spacing of consecutive predicted-sheet runs along the node normals of 2,000 sampled nodes per winding. Unlike a pair's own median spacing it cannot shrink when two windings share a sheet (on one PHerc. 0175A pair the own median fell to 1.4 voxels against a prediction pitch of 15.5).

3.3 Per-node verdicts

A node is on a sheet when a predicted-sheet voxel lies within 2 voxels (exact distance). With both ends of the node-to-neighbour segment on a sheet, the number of sheet runs along the segment gives the verdict: two runs, adjacent wraps; one run with the neighbour closer than half a pitch, the same sheet (a fit error); one run about a pitch away, the prediction fills the gap (a model error); three or more, a skipped wrap.

3.4 Windings per sheet along radial rays

In one z-slice, 180 rays are cast from the centre of the innermost winding's outline. Along each ray the crossings of every winding (its slice outline) and of every predicted sheet (runs closer than 4 voxels joined) are counted between the crossings of w040 and w115. The ratio of winding steps to sheet steps is 1 for a fit with one winding per sheet. Unlike spacing measured in radius, the count does not depend on the angle between the rays and the sheets.

3.5 Repair and pieces

Repair is a snap bounded by the neighbours: along the normal, oriented from winding $k-1$ to $k+1$, a node off any sheet, or on a neighbour's sheet closer than half a pitch, moves to the nearest sheet run strictly between the two neighbours' runs (at most 1.25 pitches); a move larger than 0.4 pitch must agree with two neighbouring moves. **Pieces** joins the on-sheet nodes of a winding across grid edges that stay on one sheet (normal step under 0.35 pitch, quarter points on a sheet), cuts at the edge of shared-sheet regions, and drops, from the inside out, the nodes that duplicate the inner neighbour's kept surface. The result is a mask per winding.

3.6 CT checks

The CT is read from existing renders (layer k lies $k-32.5$ voxels along the outward normal) and from cached level-0 chunks, never re-downloaded. Three checks are used: mean depth profiles of nodes grouped by their distance to the prediction; the CT's own sheet period along node normals (spectral peak and first autocorrelation peak); and the texture correlation between the rendered surfaces of two consecutive windings at matched nodes.

4. Results

4.1 Audit

band	stage	area	off any sheet	within 0.3 pitch of next	adjacent	same sheet	gap filled	skipped	close pairs
PHerc. 343 w047→w048 (control)	flatten	7 cm²	28 %	22 %	61 %	32 %	4 %	2 %	0 / 1
PHerc. 0175A z 6900–7900	fit (before snap)	645 cm²	46 %	3 %	87 %	6 %	4 %	3 %	50 / 119
PHerc. 0175A z 6900–7900	snap	645 cm²	14 %	21 %	71 %	26 %	1 %	2 %	42 / 119
PHerc. 0175A z 6900–7900	flatten	853 cm²	15 %	21 %	71 %	25 %	1 %	2 %	41 / 119
PHerc. 0175A z 6900–7900	flatten + repair	853 cm²	13 %	20 %	73 %	24 %	1 %	2 %	40 / 119
PHerc. 0813 z 12200–13200	snap	702 cm²	32 %	17 %	60 %	27 %	6 %	7 %	27 / 119
PHerc. 0813 z 12200–13200	flatten	1090 cm²	34 %	17 %	60 %	27 %	6 %	7 %	29 / 119
PHerc. 0800 z 5300–6300	snap	661 cm²	25 %	25 %	53 %	37 %	7 %	3 %	77 / 119
PHerc. 0800 z 5300–6300	flatten	1037 cm²	28 %	24 %	52 %	37 %	8 %	3 %	87 / 119

Table 1. Audit of each band. "Off any sheet" and "within 0.3 pitch" are fractions of all assessed nodes; the four verdicts are fractions of nodes with both ends on a sheet; close pairs have a median spacing under 0.75 of the prediction's sheet pitch. Before the snap the nodes are not yet on sheets, so the fit row's off-sheet figure is expected to be high.

On all three scrolls a large share of each flattened band lies on a neighbour's sheet or on no sheet. The overlap is partial: on PHerc. 0175A, 44 of 119 pairs share a sheet on a quarter or more of their nodes, 21 on half or more and 6 on 70 % or more; on PHerc. 0800, 55, 2 and 0.

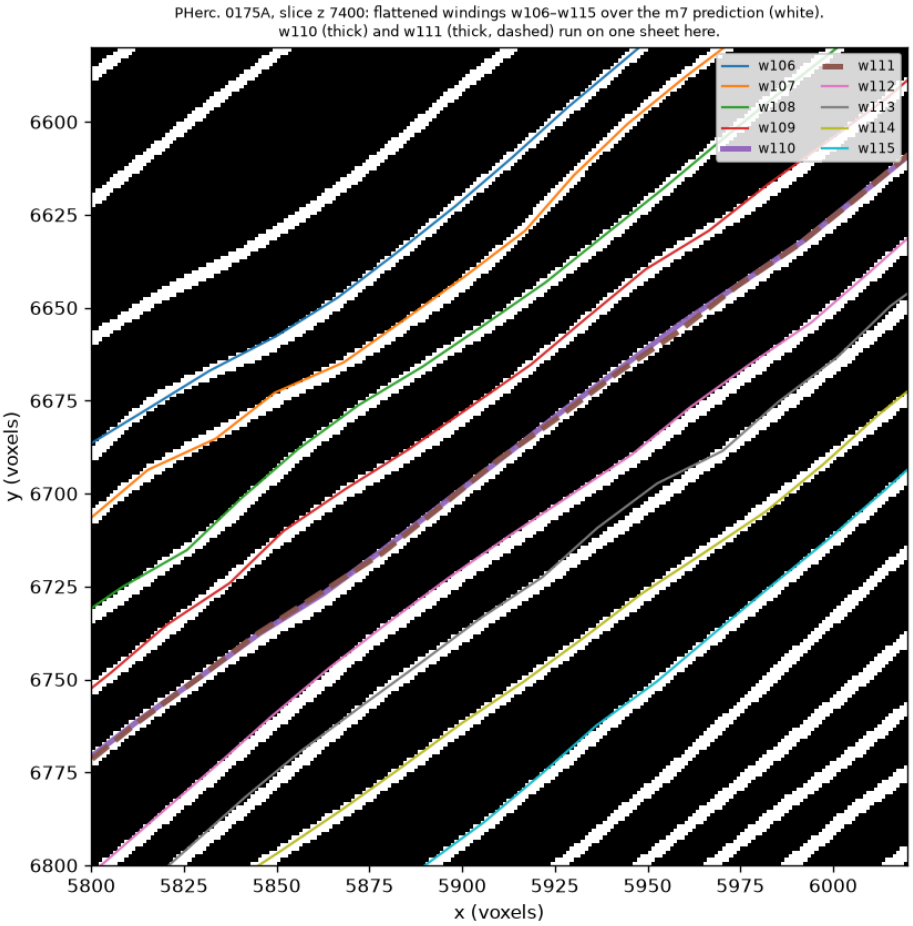


Figure 1. PHerc. 0175A, slice z 7400: flattened windings w106–w115 over the m7 prediction (white). w110 (thick) and w111 (thick, dashed) run on one sheet while each other winding has a sheet of its own.

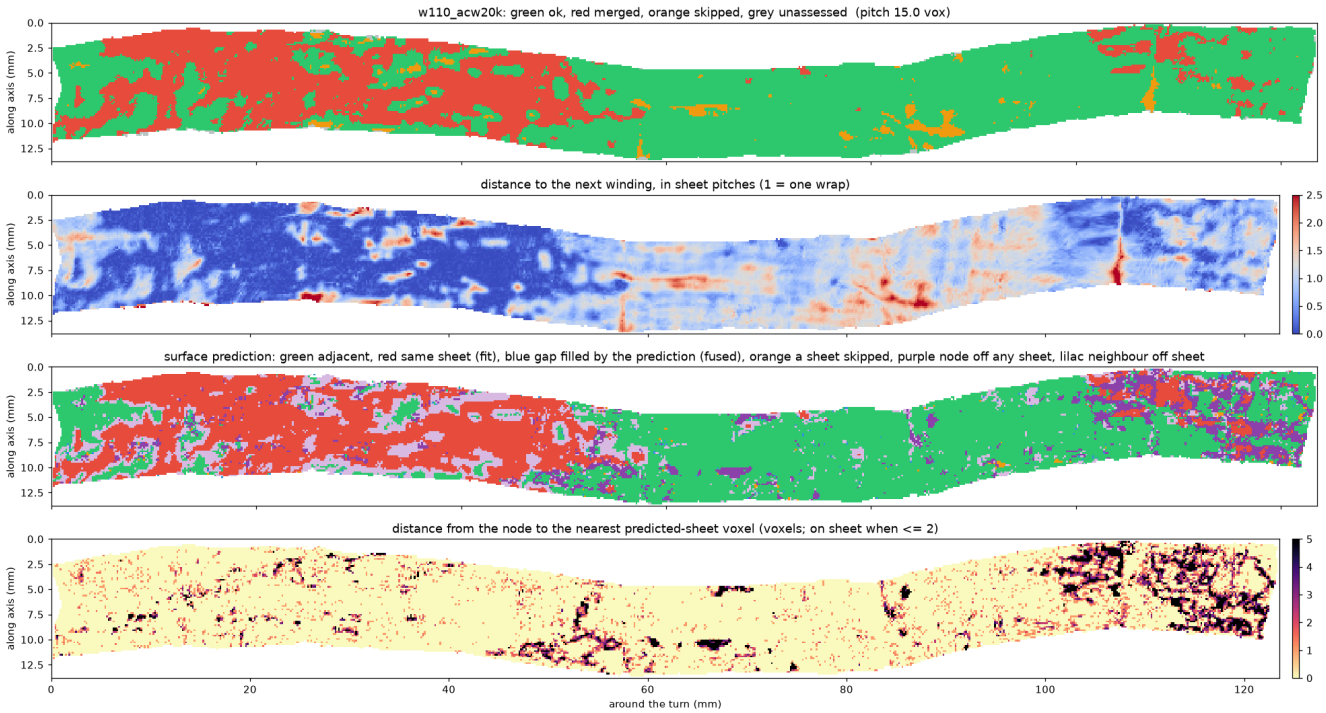


Figure 2. w110→w111 on the winding grid: spacing flags; distance to the next winding in pitches (dark blue: on top of each other); sheet verdict; node-to-sheet distance. The first 40 % of the turn shares w111's sheet.

4.2 What the CT says

Off-sheet nodes are gaps. On the PHerc. 343 control, nodes on a predicted-sheet voxel have a sharp papyrus peak 2.5 voxels outward, inside the ink window; nodes with no predicted sheet within 6 voxels sit in a CT trough about 20 grey levels darker (Figure 3).

Shared sheets are the same papyrus. On two consecutive PHerc. 0800 windings (w095, w096; Figure 4), the rendered surface patches at matched nodes correlate at a median of 0.55 where the audit puts the windings within 0.3 pitch, against 0.28 for adjacent wraps and 0.31 for random pairs (w096/w097: 0.60).

The prediction misses some sheets (correction, 13 October). Along node normals the CT's dominant period is 15–17 voxels on three PHerc. 0175A windings and 17.1–17.7 on six PHerc. 0800 windings, close to the prediction's run spacing on the same lines (14–15 and 16.2–17.5), while the fitted windings there are 9.5–13 voxels apart. A first version of this paper concluded from this that the prediction does not miss sheets. Counting bright layers along radial rays in the raw L0 CT (intensity peaks at prominence 0.5 / 1 / 1.5 local standard deviations) does not support that:

fit, slice	windings per predicted sheet	windings per CT layer	predicted / CT layers
PHerc. 0800 z 5300–6300, z 5820 (24 rays)	1.55	0.90 / 1.12 / 1.33	0.72
PHerc. 0175A 190 windings z 5900–6900, z 6460 (36 rays)	1.47	1.09 / 1.31 / 1.54	0.89

Table 3b. Raw-CT layer counts along radial rays, w040–w115. A peak count is a bound, not a count of windings: a sheet's two layers can show as two lines and two windings can touch (villa `SCROLL_ANATOMY.md`).

On PHerc. 0800 the prediction has about three quarters of the CT's bright layers, so most of the 1.5 windings per predicted sheet are sheets it misses; on PHerc. 0175A it is some of each. Figure 3b shows both: windings running through empty gaps, and bright layers with no prediction on them.

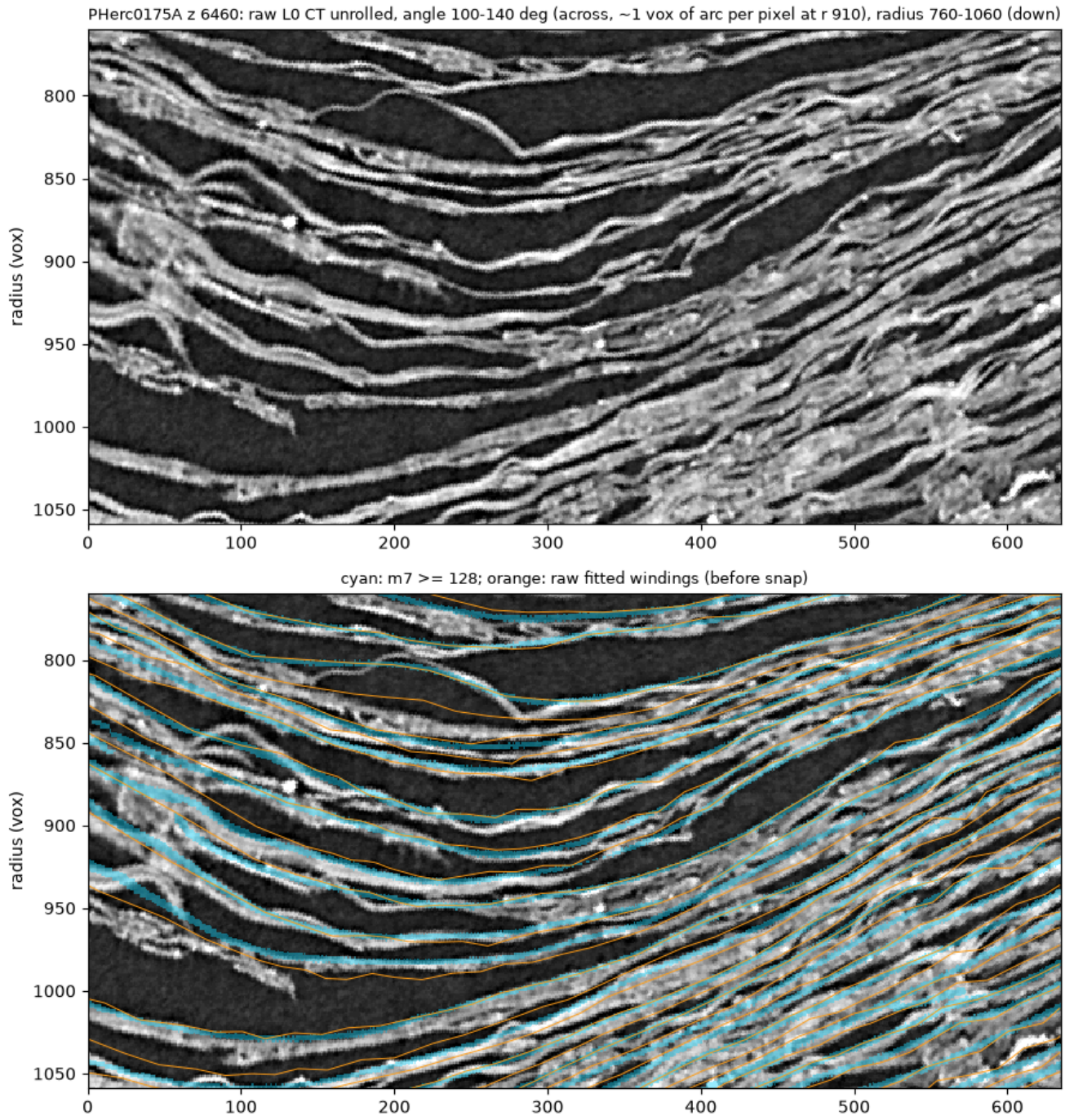


Figure 3b. PHerc. 0175A z 6460, raw L0 CT unrolled (angle 100–140°, radius 760–1060), alone and with the prediction (cyan) and the raw fitted windings (orange).

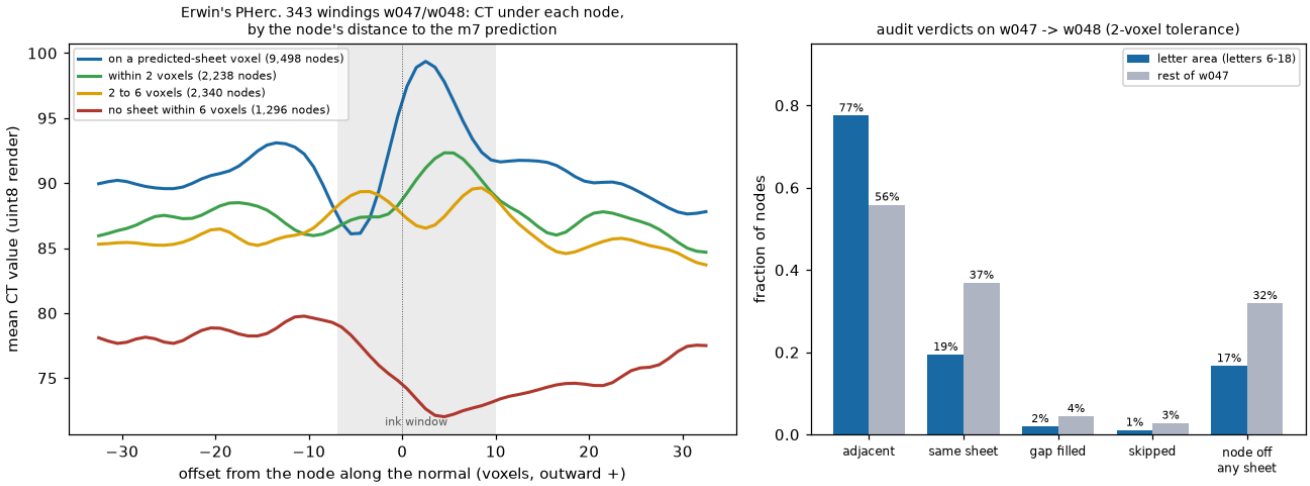


Figure 3. PHerc. 343 control (E. Nieuwlaar's w047/w048). Left: mean CT depth profile by distance to the prediction. Right: verdicts inside the letter area of w047 and on the rest of the winding.

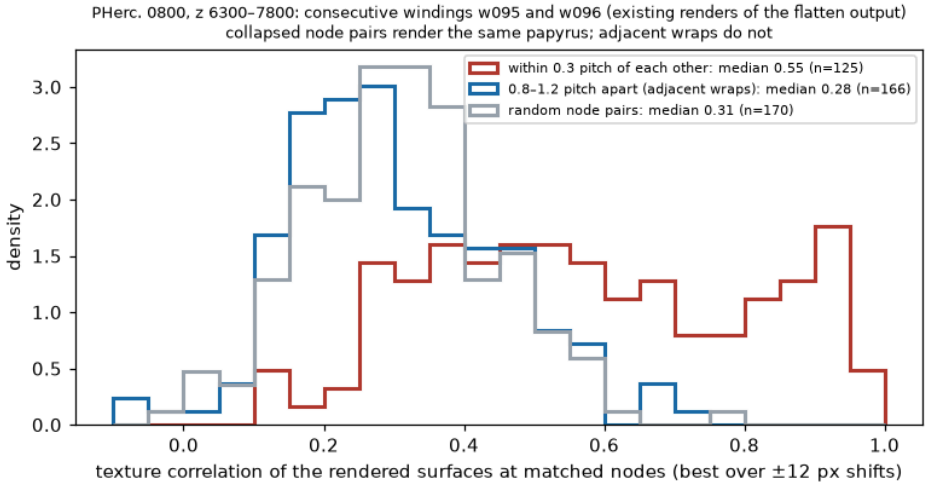


Figure 4. Texture correlation of the rendered surfaces of w095 and w096 at matched nodes.

4.3 Root cause: more windings than predicted sheets

fit, w040 to w115	slice	windings crossed	sheets crossed	windings per sheet	radius per winding
PHerc. 0800 z 6300–7800	z 6600	76	50	1.54	17.5 vox
PHerc. 0800 z 6300–7800	z 7500	76	50	1.52	19.0 vox
PHerc. 0800 z 5300–6300	z 5500	76	48	1.60	17.0 vox
PHerc. 0800 z 5300–6300	z 6100	76	52	1.47	17.1 vox
PHerc. 0175A z 6900–7900	z 7100	76	55	1.39	18.3 vox
PHerc. 0175A z 6900–7900	z 7700	76	55	1.39	18.6 vox
PHerc. 0813 z 12200–13200	z 12400	76	62	1.23	18.0 vox
PHerc. 0813 z 12200–13200	z 13000	76	63	1.20	18.1 vox

Table 2. Windings and predicted sheets crossed by 180 radial rays between the crossings of w040 and w115, before any snap (medians over rays).

Every fit advances 17–19 voxels of radius per winding, whatever the scroll, while the predicted sheets are 22–29 voxels apart along the same rays (the CT's bright layers are closer, Table 3b). `fit_spiral` learns its spiral

pitch from `model_initial_dr_per_winding` (default 16); in the PHerc. 0800 z 6300–7800 checkpoint it ended at 16.8, and in a 190-winding refit of PHerc. 0175A it read 15.6 after 3,000 iterations. Before the snap the windings do not touch (2.4 % of PHerc. 0175A nodes within 0.3 pitch), but they are 0.6–0.7 of a sheet pitch apart; the nearest-sheet snap then puts pairs of windings on one sheet over part of the turn (21 %) and leaves others to switch sheets around it. Raising the outer winding index only extends how far the fit reaches at the same pitch (the outer-shell loss is off in these configurations).

4.4 What a re-snap can and cannot repair

The neighbour-bounded repair moves 3.1 % of the PHerc. 0175A nodes. On the three rendered windings the 811 moved nodes go from a mean CT of 83 at their old position (the gap) to 104 at the new one (nodes already on their sheet: 99; Figure 5). Nodes off any sheet drop from 15.2 % to 12.7 %. Windings that share a sheet stay: with both neighbours on the sheets around it, a winding has no free sheet to move to.

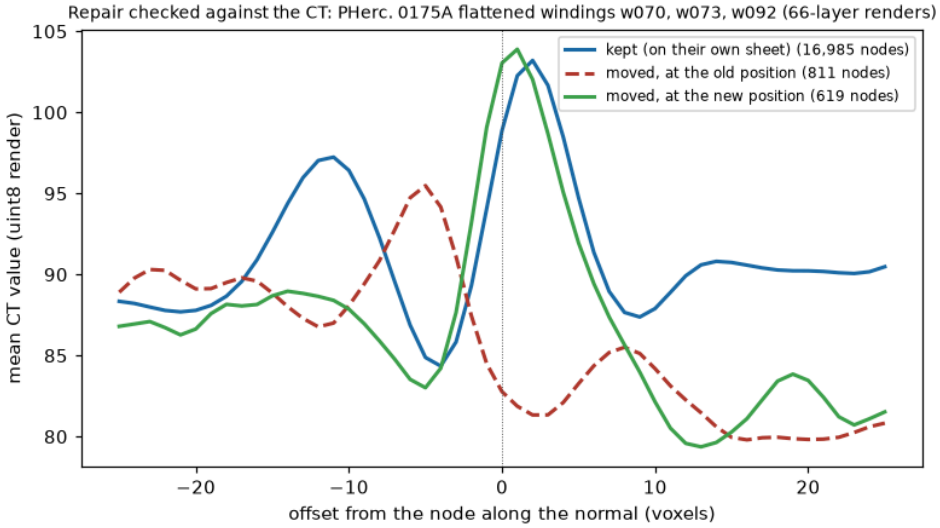


Figure 5. Mean CT depth profile of kept nodes and of moved nodes at their old and new positions.

4.5 Single-sheet pieces

band	area	on a sheet	duplicate	kept: single sheet, read once	pieces (≥ 1 cm²)
PHerc. 343 w047 + w048 (control)	14 cm²	74 %	1 cm² (6 %)	**7 cm² (48 %)**	17 (2)
PHerc. 0175A, flatten	867 cm²	85 %	118 cm² (14 %)	**513 cm² (59 %)**	873 (104)
PHerc. 0175A, flatten + repair	867 cm²	87 %	115 cm² (13 %)	**538 cm² (62 %)**	840 (107)
PHerc. 0813, flatten	1107 cm²	65 %	57 cm² (5 %)	**383 cm² (35 %)**	1345 (89)
PHerc. 0800, flatten	1054 cm²	72 %	103 cm² (10 %)	**339 cm² (32 %)**	1577 (42)

Table 3. Single-sheet pieces of each flattened band (all 120 windings).

The mask keeps papyrus (mean CT 99 at kept nodes) and drops gaps (90) and second copies (114). On the PHerc. 343 control, 15 of the 18 letter centres lie in a kept piece (one through w047's copy of the same papyrus) and the other three keep 40–64 % of their surroundings.

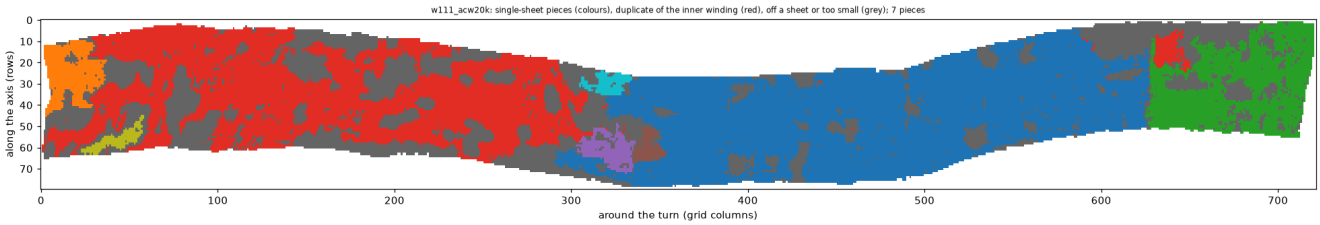


Figure 6. PHerc. 0175A w111: single-sheet pieces (colours), the part that duplicates w110 (red) and nodes off a sheet or in pieces too small to keep (grey).

5. Correcting the fits

The fits have more windings than the prediction has sheets, and the nearest-sheet snap turns that into stacked windings. Where the CT confirms the extra windings, the fit is at fault; where the prediction misses sheets (most of PHerc. 0800, Table 3b), the snap and the prediction are. Three things follow.

5.1 A check of the fit itself, before any export

A fit is a transform from scan space to a spiral space in which winding k is the curve $r = dr\,(k + \varphi/2\pi)$. Mapping the prediction's sheet voxels of a few slices through the checkpoint gives each voxel a fractional winding number w, and a correct fit puts every sheet on a whole w. The check reports, overall and per 30° sector, the phase alignment of the sheets (mean resultant length of $e^{2\pi i w}$; 1 = all on windings), the median number of windings between consecutive sheets at one angle, and the share of winding cells with no sheet nearby. It reads only the checkpoint and the prediction and runs on CPU in seconds per slice.

fit (w040–w115)	slice	windings per sheet	alignment	empty winding cells
PHerc. 0800 z 6300–7800, 20k steps	z 6600	1.53	0.14	73 %
	z 7000	1.52	0.15	71 %
	z 7500	1.54	0.11	74 %
PHerc. 0175A z 5900–6900, 190 windings, iteration 6,000	z 6200, 6600	1.42–1.44	0.02–0.05	74–75 %
same fit, new outer windings w130–w185	z 6200, 6600	1.17–1.18	0.01–0.02	81–82 %

Table 4. Spiral-space check of two fits. The PHerc. 0800 value agrees with the independent count along radial rays in the slice (1.52–1.54, Table 2).

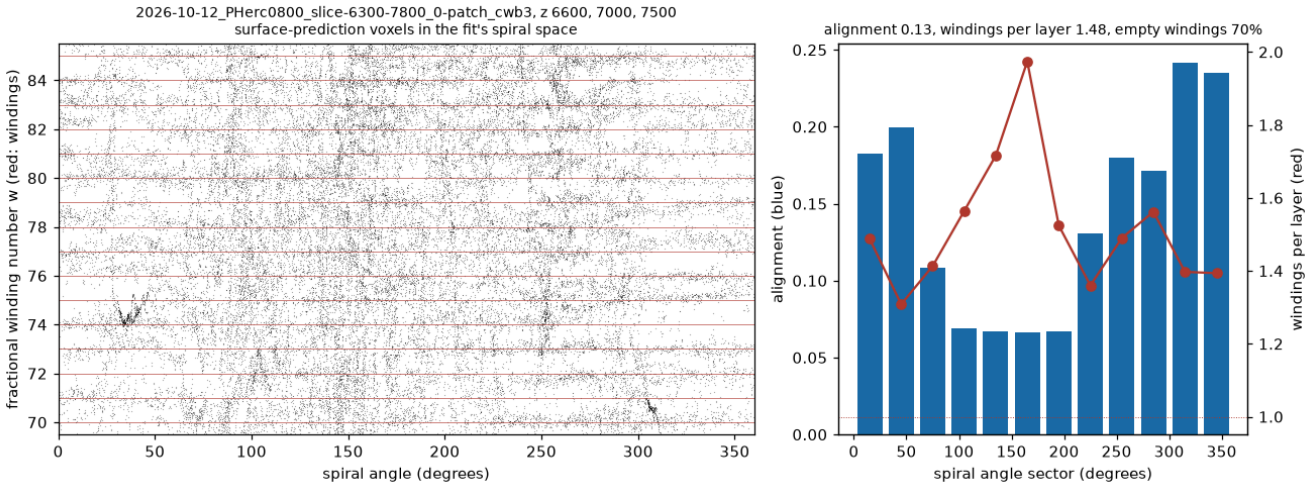


Figure 7. PHerc. 0800 z 6300–7800 fit: the prediction's sheet voxels of three slices in the fit's spiral space (red lines: windings) and, per sector, the alignment and the windings per sheet. Over 90–210°, where the prediction is fused, the fit's angle field crosses the sheets and the alignment falls to 0.07 (0.01–0.05 in single slices).

5.2 What a re-snap can and cannot do

A snap moves nodes onto existing sheets. A fit with 1.5 windings per predicted sheet has half again as many windings as the prediction has sheets in that range, so after a snap onto the prediction a third of its windings must share a sheet with a neighbour or lie in a gap; the neighbour-bounded repair (Section 4.4) fixes the nodes that have a free sheet to go to, and the pieces mask (4.5) marks the rest. Raising the outer winding index does not help by itself: in these configurations the outer-shell loss is off, the index only extends how far the fit reaches, and the 190-winding refit, at iteration 6,000, still reads 1.4 inside and 1.2 for the new outer windings (Table 4).

5.3 What should change in the fit

1. **Start from the roll's own pitch, measured in the CT.** `fit_spiral` moves its pitch little from the starting value (16 → 16.8 voxels), so the starting value should be the scroll's sheet spacing. The prediction's spacing (25–29 voxels on PHerc. 0800) overstates it where the prediction misses sheets; the raw CT's bright layers are about 18–23 voxels apart along rays there. A snap that never puts two windings on one predicted surface, and better prediction recall, address the rest.
2. **Use the alignment as a loss.** The phase alignment of the prediction's sheet voxels is differentiable in the transform and the pitch; a term rewarding whole winding numbers would pull windings onto sheets and push empty windings out of the gaps.
3. **Check every fit before rendering.** The spiral-space check and the density count take seconds on CPU; a fit that reads more than about 1.1 windings per sheet will produce duplicated papyrus after the snap.

Items 1 and 2 need GPU refits to confirm; this work ran on CPU only, so they are stated as the change the measurements call for, not as a demonstrated result.

6. Integration

- **Standalone tools** (MIT, 23 tests on a synthetic scroll with injected faults): `density`, `audit`, `repair`, `pieces`; tifxyz and OME-Zarr in, tifxyz, CSV/JSON and figures out. github.com/TAUIL-Abd-Elilah/spiral-winding-audit
- **In villa:** `spiral-fitting/check_fit_against_prediction.py` with a regression test, on branch `tauil/spiral-sheet-reindex` of the author's fork, proposed to ScrollPrize/villa.

- **Where they sit in the recipe:** run `check_fit_against_prediction` or `density` right after the fit; `audit` after the snap and after flattening; `repair` before flattening; `pieces` to choose what to render and read.

7. Limits

- Agreement with m7 is agreement with a model. The CT checks cover the control pair, three rendered PHerc. 0175A windings and seven PHerc. 0800 windings.
- The neighbour's surface is its tangent plane at the nearest node, not the exact mesh (agreement within one voxel for about 90 % of nodes).
- The prediction pitch is a median over sampled nodes; missed sheets lengthen it. Along radial rays the prediction has 72–89 % of the CT's bright layers (Table 3b), so pitch-relative numbers overstate the extra windings.
- The pieces mask is conservative: it drops some papyrus where the prediction has holes along the sheet.

8. Reproducing

```
pip install -e .
winding-audit density <fit>/meshes/fitted_<tag> --surface <m7.zarr> --z 6600 7500 --first w040 --last w115
winding-audit audit   <band> --surface <m7.zarr> --out runs/band --voxel-um 8.64 [--subdir
tifxyz/flatten.tifxyz]
winding-audit repair  <band> --surface <m7.zarr> --out runs/band_repaired --audit --voxel-um 8.64
winding-audit pieces  <band> --surface <m7.zarr> --out runs/band_pieces --voxel-um 8.64 --figures
```

All result tables are in `docs/results/` of the repository; the test suite builds a synthetic spiral scroll with injected mergers, skips, gaps and fused prediction and checks every verdict.

References

1. ScrollPrize/villa: `volume-cartographer/scripts/spiral` (fit_spiral) and lasagna. github.com/ScrollPrize/villa
2. E. Nieuwlaar, PHerc. 343 First Letters (recipe, snap_meshes.py, released windings). github.com/Nieuwlaar/pherc343-first-letters
3. Vesuvius Challenge open data: m7 surface predictions and CT volumes. scrollprize.org/data
4. Vesuvius Challenge, Progress Prizes and Open Problems. scrollprize.org/prizes